

Bats

PHYLUM	Chordata
CLASS	Mammalia
ORDER	Chiroptera
FAMILIES	18
SPECIES	1,330

CLASSIFICATION NOTE

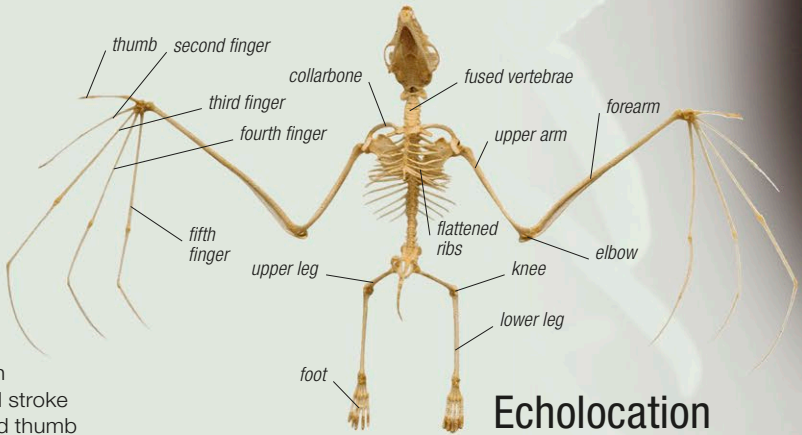
Chiroptera has 2 suborders: Megachiroptera, comprising one family (Pteropodidae), and Microchiroptera (all other bats). Megachiropterans have a foxlike face that lacks features for echolocation (see below).

Bats are the only mammals that possess true, flapping wings and the ability to fly (as opposed to colugos, for example, which glide). Bats' wing membrane (the patagium), an extension of the skin of the back and belly, provides a high degree of maneuverability in flight. Wingspans range from over 5 ft (1.5 m) in the large flying fox to as little as 6 in (15 cm) in the hog-nosed bat. Over half the species echolocate (see below) to capture prey and to navigate at night. Chiroptera is a huge order that comprises nearly a quarter of all mammal species and is exceeded only by rodents in terms of species numbers. Bats are common in tropical

and temperate habitats worldwide but are not found in environments that are too cold to support a source of food, such as the polar regions.

Anatomy

Perhaps the most distinctive feature of bats is their wings, which are formed from a double layer of skin stretched between the side of the body and the 4 elongated fingers on each hand. Blood vessels and nerves run between these 2 layers. Extra support is required for the arms to be used as wings, and this is provided by features such as fused vertebrae, flattened ribs, and a strong collarbone. The sternum (breastbone) has a central ridge to which the large muscles used in the downward stroke of the wing are attached. A short, clawed thumb is present in most species at the point where the fingers join, and a cartilaginous spur (the calcar) on the inside of the ankle joint assists in spreading the tail membrane.

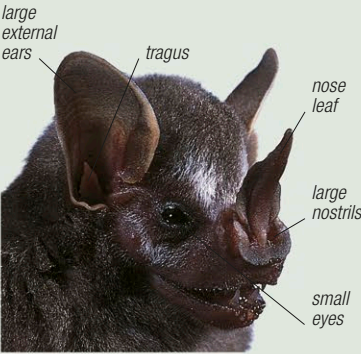


SKELETAL FEATURES

A bat's arms, legs, and greatly elongated fingers provide the framework for the wings. Bats' legs have been rotated 180 degrees so that the knee and the foot bend in the opposite direction to the knee and back foot of other mammals.

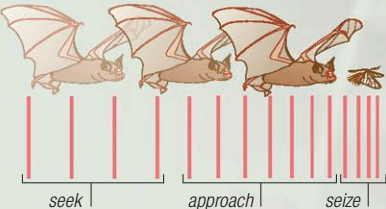
Echolocation

All microchiropterans echolocate. When used in flight, this navigation system makes them formidable hunters. Sounds ("clicks") are produced in the larynx, emitted through the nose or mouth, and directed or focused by the nose leaf (if present). Once the clicks have reflected off an object, the returning echo is picked up via the bat's sensitive ears. The time it takes to receive the echo reveals the size and location of anything in the bat's path.



EQUIPPED FOR ECHOLLOCATION

Although bats' eyes are well developed, hearing and the sense of smell are more important than sight. Many microchiropterans, such as this neotropical fruit bat, also have a large nose leaf, which assists echolocation. The function of the tragus (a lobe in the front of the ear) is uncertain: it may improve the accuracy of echolocation. Megachiropteran bats generally do not echolocate and have larger eyes (for detecting prey) and smaller ears.



PHASES OF ECHOLLOCATION AS A BAT HUNTS AN INSECT

USING SOUND TO FIND PREY

All of the insect-eating bats use echolocation to find airborne prey. When searching for food, such as mosquitoes and moths, the bat emits a series of clicks, represented by red bars on this diagram. As the bat approaches its prey, the time between "clicks" shortens. This helps the bat to pinpoint its target.

CATCHING PREY FROM THE AIR



CLOSING IN ON THE PREY

Using echolocation, this fisherman bat has pinpointed a small surface-swimming fish.

CAPTURE

In one swift movement, the bat catches the fish by raking the water with its long, sharp claws.

HOLDING ON

The bat quickly transfers the fish from its claws to its mouth, so that the fish cannot escape.

FINDING A PLACE TO FEED

In order to eat the fish, the bat must first locate a tree on which to land.